EVERY INVERTIBLE SHEAF ON THE SOURCE OF A FINITE MORPHISM THAT IS AN ISOMORPHISM OUTSIDE A DISCRETE SET IS PULLED BACK FROM THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Picard group as H^1 of the unit sheaf and the Leray spectral sequence. Stacks Project, Cohomology, Lemmas cohomology-lemma-h1-invertible (tag 09NU) and cohomology-lemma-Leray (tag 01F2). Convert the Picard statement into a statement about H^1 and provide the spectral sequence comparison.
- Trivializing invertible sheaves after a finite morphism and cohomology supported on a closed subset. Stacks Project, Divisors, Lemma divisors-lemma-finite-trivialize-invertible-upstairs (tag 0BUT); Cohomology, Lemma cohomology-lemma-cohomology-and-closed-immersions (tag 02UV). The first kills the edge map and the second handles sheaves supported on the discrete complement.

Source reference. Original reference 09NU: sources/stacks-chow/tags.txt, source lines 12227–12227. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01F2: sources/stacks-chow/tags.txt, source lines 1855–1855. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0BUT: sources/stacks-chow/tags.txt, source lines 14886–14886. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02UV: sources/stacks-chow/tags.txt, source lines 3746–3746. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01AX: sources/stacks-chow/tags.txt, source lines 1706–1706. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a finite morphism of schemes. Assume there exists an open $V \subset Y$ such that $f^{-1}(V) \to V$ is an isomorphism and $Y \setminus V$ is a discrete space. Then every invertible $\mathcal{O}_X$-module is the pullback of an invertible $\mathcal{O}_Y$-module.*

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. We will use that $\mathrm{Pic}(X) = H^1(X, \mathcal{O}_X^*)$, see Cohomology, Lemma 09NU. Consider the Leray spectral sequence for the abelian sheaf $\mathcal{O}_X^*$ and f, see Cohomology, Lemma 01F2. Consider the induced map

$$H^1(X, \mathcal{O}_X^*) \longrightarrow H^0(Y, R^1 f_* \mathcal{O}_X^*)$$

Divisors, Lemma 0BUT says exactly that this map is zero. Hence Leray gives $H^1(X, \mathcal{O}_X^*) = H^1(Y, f_* \mathcal{O}_X^*)$. Next we consider the map

$$f^\sharp : \mathcal{O}_Y^* \longrightarrow f_* \mathcal{O}_X^*$$

By assumption the kernel and cokernel of this map are supported on the closed subset $T = Y \setminus V$ of Y. Since T is a discrete topological space by assumption the higher cohomology groups of any abelian sheaf on Y supported on T is zero (follows from Cohomology, Lemma 02UV, Modules, Lemma 01AX, and the fact that $H^i(T, \mathcal{F}) = 0$ for any $i > 0$ and any abelian sheaf $\mathcal{F}$ on T). Breaking the displayed map into short exact sequences

$$0 \to \mathrm{Ker}(f^\sharp) \to \mathcal{O}_Y^* \to \mathrm{Im}(f^\sharp) \to 0, \quad 0 \to \mathrm{Im}(f^\sharp) \to f_* \mathcal{O}_X^* \to \mathrm{Coker}(f^\sharp) \to 0$$

we first conclude that $H^1(Y, \mathcal{O}_Y^*) \to H^1(Y, \mathrm{Im}(f^\sharp))$ is surjective and then that $H^1(Y, \mathrm{Im}(f^\sharp)) \to H^1(Y, f_* \mathcal{O}_X^*)$ is surjective. Combining all the above we find that $H^1(Y, \mathcal{O}_Y^*) \to H^1(X, \mathcal{O}_X^*)$ is surjective as desired. $\square$